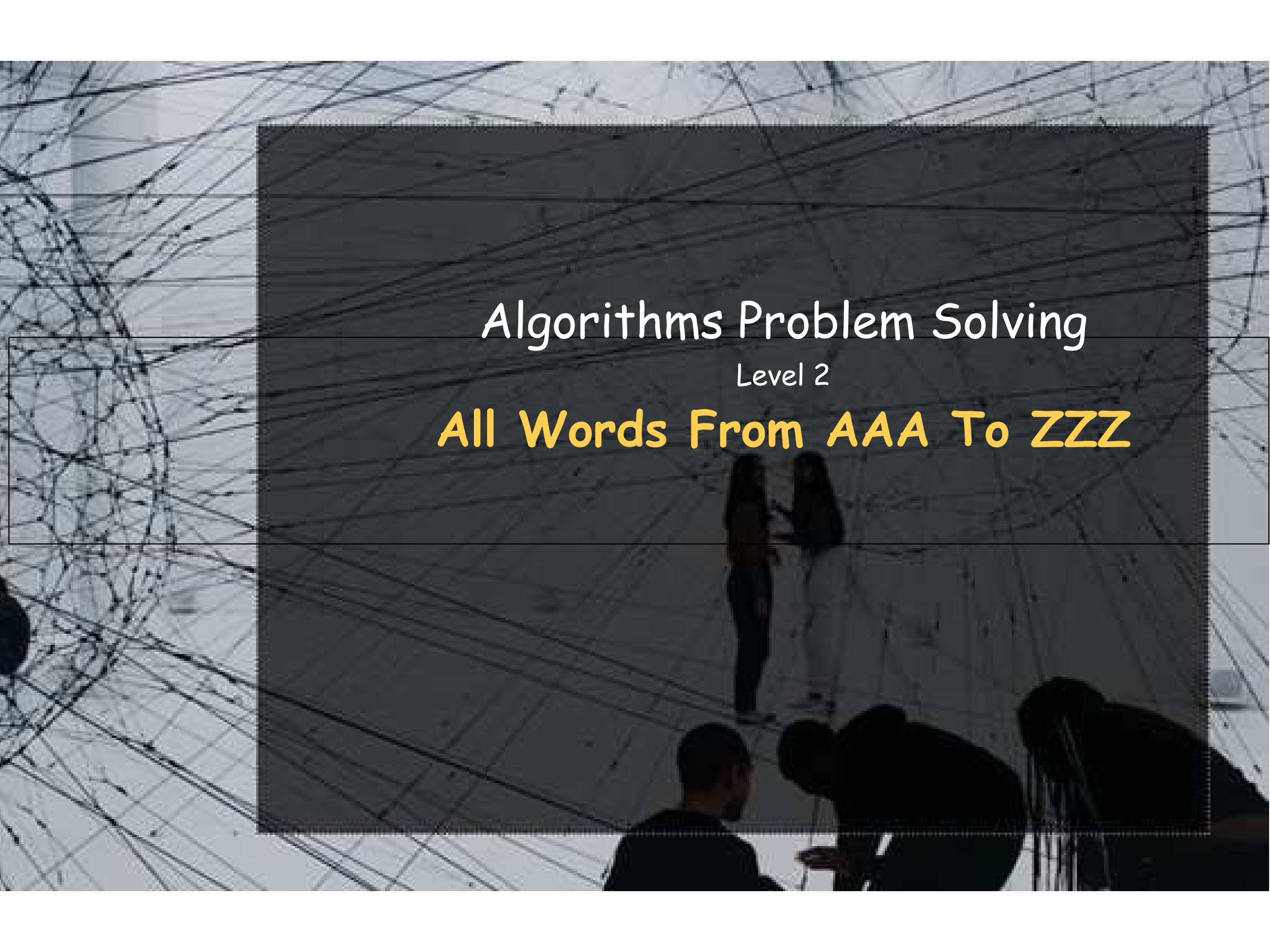


Algorithms Problem Solving Level 2

All Words From AAA To
ZZZ



Algorithms Problem Solving

Level 2

All Words From AAA To ZZZ

Definition

- ◆ A program that prints all possible three-letter combinations from *AAA* to *ZZZ*.
- ◆ Each letter can be any uppercase letter from *A* to *Z*.

Explanation

- ◆ The program uses three nested `for` loops to generate the combinations.
- ◆ The first loop controls the first letter `i`.
- ◆ The second loop controls the second letter `j`.
- ◆ The third loop controls the third letter `k`.
- ◆ `cout << i << j << k` prints the three letters together.
- ◆ Each loop runs from `A` to `Z`, giving 26 possible values.
- ◆ Therefore, the total number of combinations is $26 \times 26 \times 26 = 17,576$.

Steps in main()

1. Start the first loop from `A` to `Z`.
2. For each value of `i`, run the second loop from `A` to `Z`.
3. For each value of `j`, run the third loop from `A` to `Z`.
4. Print `i`, `j`, and `k` together.
5. Repeat until all combinations from `AAA` to `ZZZ` are printed.
6. Print a separator after completing all combinations for each first letter.

💡 Example

- ◆ When `i = A`, the program prints all combinations from `AAA` to `AZZ`.
- ◆ When `i = B`, the program prints all combinations from `BAA` to `BZZ`.
- ◆ The process continues until `i = Z`.

➡ Result

- ◆ The output starts with:

AAA

AAB

AAC

AAD

...

ZZW

ZZX

ZZY

ZZZ

Note

- ◆ The three nested loops work together to generate every possible combination.
- ◆ For every value of `i`, the complete `j` and `k` loops run again.
- ◆ The separator `-----` is printed after all combinations beginning with the same first letter have been completed.

⊘ Warning

- ◆ The comment `// Function to make the inverted Number` is incorrect because the function generates letter combinations, not an inverted number.
- ◆ The program assumes uppercase English letters from `A` to `Z`.

Comparison

- ▣ One loop → prints one sequence, such as `A` to `Z`.
- ▣ Two nested loops → can generate two-letter combinations such as `AA`, `AB`, `AC`, ..., `ZZ`.
- ▣ Three nested loops → generates three-letter combinations such as `AAA`, `AAB`, `AAC`, ..., `ZZZ`.

✦ Summary

- ✓ Three nested loops generate all three-letter combinations.
- ✓ Each loop runs from `A` to `Z`.
- ✓ `i` controls the first letter.
- ✓ `j` controls the second letter.
- ✓ `k` controls the third letter.
- ✓ Total combinations = $26 \times 26 \times 26 = 17,576$.

Thank you 😊